Exercise Sheet 5

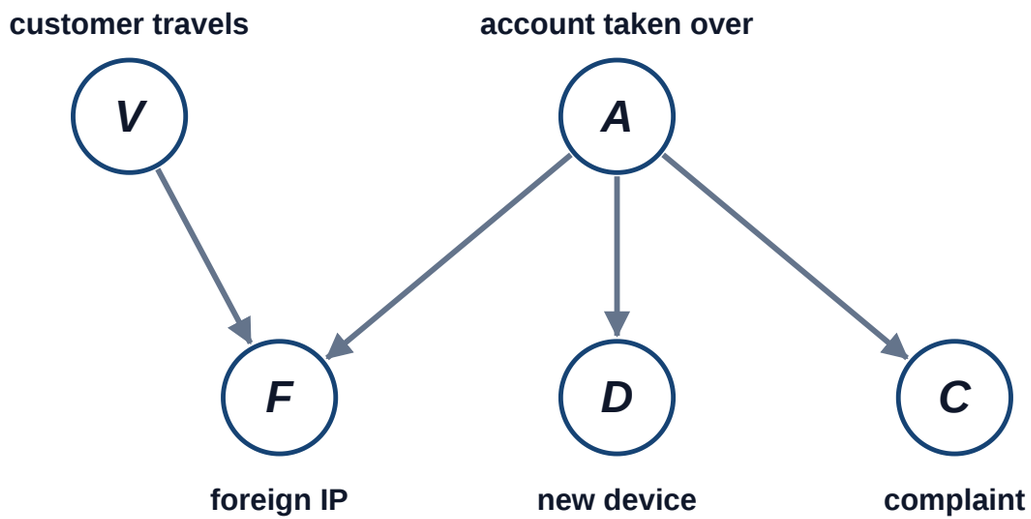
Bayesian Networks

Probabilistic Machine Learning

Prof. Dr. Michael Granitzer

5.A Graphing a Joint

Exercise 5.A.1: A Small Fraud Network



All five variables are yes/no.

1. Write the joint $p(V, A, F, D, C)$ as the network factorises it.
2. How many numbers does the network need? How many the full joint?
3. State the local Markov claim for D and for F.
4. Name one missing arrow you would challenge, and what adding it costs.

Exercise 5.A.2: Inference by Elimination

Tables: $P(V{=}1) = 0.05$, $P(A{=}1) = 0.01$; $P(F{=}1 \mid V, A)$: 0.02 for $(0,0)$, 0.9 for $(1,0)$, 0.7 for $(0,1)$, 0.95 for $(1,1)$; $P(D{=}1 \mid A)$: 0.1 for $A{=}0$, 0.6 for $A{=}1$; $P(C{=}1 \mid A)$: 0.001 for $A{=}0$, 0.5 for $A{=}1$.

1. Compute $P(F{=}1)$. Which variables can be dropped before summing, and why? Eliminate V first and give the message $m(A)$.
2. Compute $P(A{=}1 \mid F{=}1)$.
3. Compute $P(A{=}1 \mid F{=}1, D{=}1)$.
4. Compute $P(A{=}1 \mid F{=}1, V{=}1)$ and compare with 2. Name the effect.

5.B Reading Independence

Exercise 5.B.1: d-Separation

For the fraud network of 5.A.1, decide with d-separation — give the path and the triple that blocks or opens it:

1. $V \perp\!\!\!\perp A$?
2. $V \perp\!\!\!\perp A \mid F$?
3. $V \perp\!\!\!\perp D$?
4. $V \perp\!\!\!\perp D \mid F$?
5. $V \perp\!\!\!\perp D \mid F, A$?
6. $D \perp\!\!\!\perp C$? and $D \perp\!\!\!\perp C \mid A$?

Then check 3 and 4 with the numbers of 5.A.2: compare $P(V=1 \mid F=1)$ with $P(V=1 \mid F=1, D=1)$.

Exercise 5.B.2: Which Arrows Are in the Data?

1. Which of these graphs are Markov equivalent? (a) $X \to Y \to Z$ · (b) $X \leftarrow Y \leftarrow Z$ · (c) $X \leftarrow Y \to Z$ · (d) $X \to Y \leftarrow Z$ · (e) (a) plus $X \to Z$
2. In the fraud network of 5.A.1, which single arrows can be reversed without changing any independence claim? Can $A \to D$ and $A \to C$ be reversed together?
3. An analyst fits the fraud network from data and says "take-overs **cause** new devices". What can the data support?

5.C Plates and Learning

Exercise 5.C.1: Naive Bayes as a Plate Diagram

Naive Bayes for N transactions with p yes/no flags: class prior $\pi = P(y = 1)$ with a $\mathrm{Beta}(a, b)$ prior; flag probabilities $\theta_{jc} = P(x_j = 1 \mid y = c)$, each with a $\mathrm{Beta}(1, 1)$ prior.

1. Draw the plate diagram: nodes, shading, dots, plates (and which plates are nested).
2. Write the factorisation that the diagram encodes.
3. At prediction time a new transaction arrives with flags but no label. What changes in the drawing?

Exercise 5.C.2: Counting, Elimination, and an Empty Row

1. A chain $X_1 \to \cdots \to X_{10}$ of yes/no variables. How many terms does $p(x_{10})$ need if all other variables are summed at once, and roughly how many operations by elimination?
2. From 200 records of the fraud network (table), estimate $P(F{=}1 \mid V, A)$ by maximum likelihood and with one pseudo-count per value. What happens to the last row, and what does smoothing put there?

V	A	records	with $F = 1$
0	0	180	4
1	0	15	13
0	1	5	3
1	1	0	0